Introduction

The business problem is setting optimal prices for a fast-paced fashion industry where demand is uncertain and interdependent across competing fashion styles. The example taken here is **Rue La La**, where they consistently have limited-time online sales. Their aim is to optimise pricing strategies to match expected demand with limited supply, ensuring minimal stockout scenarios.

In this case, Rue La La faces two major challenges:

1. Predicting demand for new products that have no historical sales data
2. Optimizing prices to manage demand based on the supply of each clothing

Models

There are 2 parts to the solution, 1 tackling the problem of demand prediction, and the other tackling the problem of price optimisation. We can use the **kNN** model & **linear regression** to predict demand, and **Neural Networks** to optimise prices and match demand & supply.

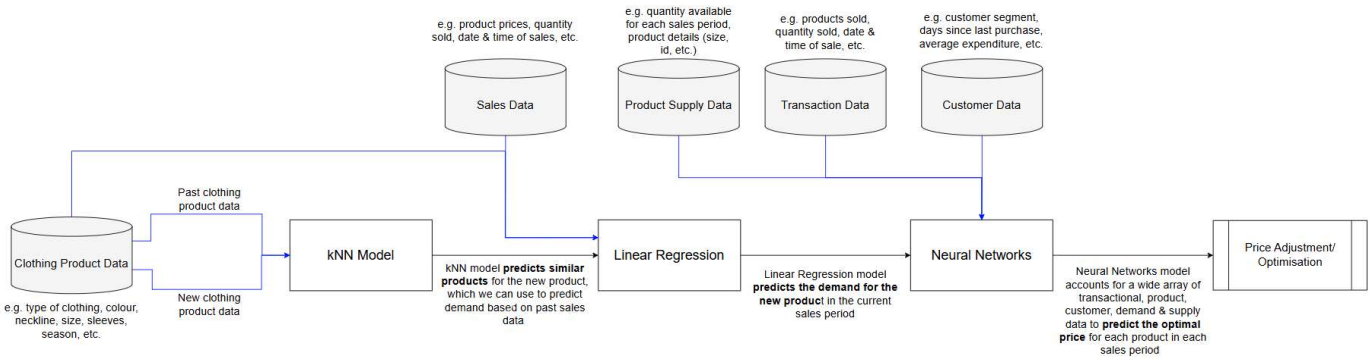
In the fast-fashion industry, the company releases new products consistently and does not have past sales data to predict the demand of the new product. Therefore, we use the kNN model to find similar past products to use as benchmarks for the new product. Thereafter, we can use linear regression using the past sales data of similar products to predict the demand for this new product. **Models used for demand prediction:**

- kNN model to find similar products for the new product
- Using the similar products, use linear regression with price data to predict demand for this new product

After identifying the potential demand of this new product, we then train a neural networks model using granular past transaction data, customer behavioural data, product supply & forecasted demand data to optimise the prices of each product. This neural networks model can dynamically adjust its pricing recommendations using new sales data during each sales period. **Models used for price optimisation:**

- Neural Networks to optimise product prices and match demand & supply
- When predicted sales deviate from the actual sales, adjust prices to stimulate/dampen demand

Model Ensemble



Data Required & Collection

For each model, we will train on different types of data.

For the **kNN** model, we will primarily be training on the **clothing product data**, such as type of clothing, colour, size, neckline, etc., to help us predict which clothes are highly similar to each other. This will help us identify the most similar products for each new product. The clothing product data can be collected when each product is entered into the catalog or database, making sure that the essential attributes of each clothing is collected.

For the **linear regression** model, we will be training on both **clothing product data** & **past sales data**. Past sales data will include variables like price sold, quantity sold, date & time of sale, etc., to help with the demand prediction for each type of product. Clothing product data will be the same as with the kNN model. Past sales data will be collected using every past transaction made by the company.

For the **neural networks** model, we will be training on **past sales transaction data**, **customer attributes**, **product supply quantities**, and **forecasted demand from linear regression**. Customer attributes can include variables like customer segments, days from last purchase, average expenditure per transaction, most purchased product, etc., to account for likely customers that will purchase a certain product. Product supply quantities are based on the available quantities for each product at a certain date. Past sales transaction data & forecasted demand are the same as with the linear regression model. Customer attributes can be collected when the customer first registers with the company and derived based on their past purchasing behaviour. Product supply quantities can be collected through supply orders with the upstream supplier.

Model Refresh Frequency

Model refresh frequency will differ from model to model.

Since the **kNN & Linear regression models** are used for the same demand prediction purpose, both models will have the same refresh frequency. Given that demand prediction is only crucial at the point of procuring supplies, we will expect to refresh the model when there are products to be procured.

For the **neural networks model**, the refresh frequency will depend on the company's pricing strategy. If prices are dynamically changed intra-day, the model should be refreshed intra-day (e.g. every hour, every 15 minutes) using new sales, demand, and supply data from intra-day sales; otherwise, if prices are only updated daily, the model should be refreshed daily using new sales, demand, and supply data from the previous day.